



Structure-performance relationship of nanodiamonds @ nitrogen-doped mesoporous carbon in the direct dehydrogenation of ethylbenzene

Yuefeng Liu^{a,b,*}, Housseinou Ba^b, Jingjie Luo^c, Kuang-Hsu Wu^c, Jean-Mario Nhut^b,
Dang Sheng Su^{a,c,**}, Cuong Pham-Huu^{b,***}

^a Dalian National Laboratory for Clean Energy (DNL), Dalian Institute of Chemical Physics, Chinese Academy of Sciences, 457 Zhongshan Road, 116023 Dalian, China

^b Institut de Chimie et Procédés pour l'Energie, l'Environnement et la Santé (ICPEES), ECPM, UMR 7515 du CNRS-Université de Strasbourg, 25 rue Becquerel, 67087 Strasbourg Cedex 02, France

^c Shenyang National Laboratory for Materials Science, Institute of Metals Research, Chinese Academy of Sciences, 72 Wenhua Road, 110016 Shenyang, China

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ABSTRACT

Nanocarbon materials have been reported as an alternative robust metal-free catalyst in the field of the catalytic dehydrogenation with improved catalytic performance as well as stability. In this study, the hybrid metal-free catalyst consisting of dispersed nanodiamonds within a nitrogen-doped mesoporous carbon (ND@NMC) phase was investigated. Such material with high effective surface area and porosity was prepared under different thermal treatment temperatures and further evaluated for the direct dehydrogenation (DDH) of ethylbenzene (EB) to styrene (ST). The characterization techniques such as N₂ adsorption-desorption, X-ray photoelectron spectroscopy, high-resolution transmission electron microscopy and Raman spectroscopic analysis were used to investigate the surface properties and structures of the as-prepared ND@NMC composites. The ND@NMC-700 catalyst annealed at 700 °C presented a ST specific reaction rate and a relative areal activity of 5.8 mmol_{ST} g_{catalyst}⁻¹ h⁻¹ and 0.28 μmol_{ST} m⁻² h⁻¹ with a ST selectivity of 99.6%, which is the highest DDH activity among the investigated nanocarbons including ND, carbon nanotubes, NMC and ND@MC (ND covered by mesoporous carbon) catalysts. The superior dehydrogenation performance could be attributed to the high dispersion of the metal-free nanodiamond centers within the NMC layer which provided a well surface contact with the reactant. It can also be confirmed that the rational contents of ketone (C=O) functional groups, as well as the opened porous network in ND@NMC-700 catalyst resulted to the superior DDH activity and styrene selectivity. Moreover, the presented nitrogen groups are beneficial for construction of surface defects and porosity as well as the improvement of styrene selectivity.

1. Introduction

Carbon nanomaterials (such as carbon nanotubes/nanofibers, graphene, nanodiamonds, and the related nanocarbons doped with heteroatoms) presenting unique structural features and chemical properties, have been widely investigated in the last decades in various important catalytic processes [1–10]. Among the series of carbon nanomaterials, the detonation nanodiamond (ND), a typical sp³-hybridized carbon nanoparticle with an average size less than 10 nm, has attracted significant interests as a catalyst support or directly as an effective metal-free catalyst [11–18]. It was reported that ND could be considered as an alternative metal-free catalyst for the production of

styrene through the direct dehydrogenation (DDH) of ethylbenzene (EB) [12]. Su and co-workers [12] have shown that nanodiamond presented superior dehydrogenation activity than other carbon nanomaterials, i.e. carbon nanotubes, nanographite and mesoporous carbon in the DDH ethylbenzene reaction to produce styrene, and exhibited DDH activity about three times higher than the one obtained with the iron-based industrial catalyst with similar selectivity towards styrene formation. When the surface of NDs is modified by nitrogen [18] or coated by a layer of carbon nitride hybrid [19,20], the catalytic activity on those modified NDs could be enhanced significantly. It has been demonstrated that the incorporation of nitrogen(s) in the carbon matrix network gives rise to an electronic modification of the adjacent carbon

* Corresponding author.

** Corresponding author at: Dalian National Laboratory for Clean Energy (DNL), Dalian Institute of Chemical Physics, Chinese Academy of Sciences, 457 Zhongshan Road, 116023, Dalian, China.

*** Corresponding author.

E-mail addresses: yuefeng.liu@dicp.ac.cn (Y. Liu), dssu@dicp.ac.cn (D.S. Su), cuong.pham-huu@unistra.fr (C. Pham-Huu).